

Course Title : Big Data Analytics

Course Code : Z1AMC 702

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Experiment 1: Hadoop Single Node Installation

1. Objective

To install and configure Hadoop in a single-node environment and verify the successful execution of Hadoop Distributed File System (HDFS) and YARN services.

2. Procedure

- Install Java Development Kit (JDK).
- Download and extract the Hadoop package.
- Configure environment variables in .bashrc.
- Configure Hadoop files:
 - core-site.xml
 - hdfs-site.xml
 - mapred-site.xml
 - yarn-site.xml
- **Format the NameNode using:**
hdfs namenode -format
- **Start HDFS services using:**
start-dfs.sh
- **Start YARN services using:**
start-yarn.sh

➤ **Verify running services using:**

jps

➤ **Access Hadoop web interfaces:**

- NameNode: <http://localhost:9870>
- ResourceManager: <http://localhost:8088>

3. Source Code

export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64

export HADOOP_HOME=/usr/local/hadoop

export PATH=\$PATH:\$HADOOP_HOME/bin:\$HADOOP_HOME/sbin

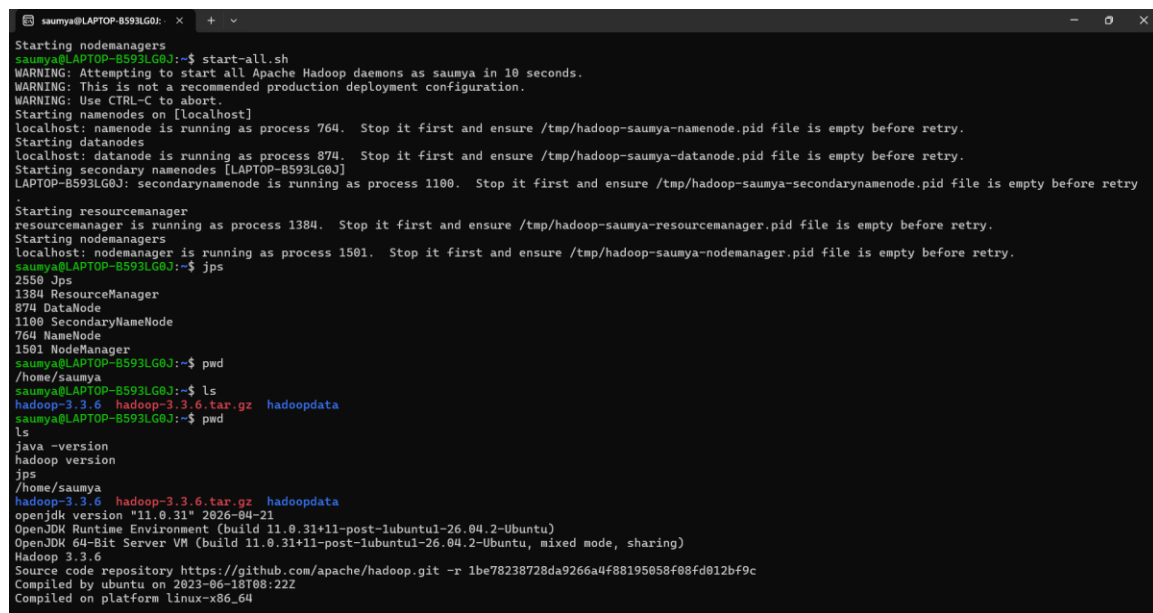
hdfs namenode -format

start-dfs.sh

start-yarn.sh

jps

4. Screen Shot



```
saumya@LAPTOP-B593LG0J: x + v
Starting nodemanagers
saumya@LAPTOP-B593LG0J:~$ start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as saumya in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [localhost]
localhost: namenode is running as process 764. Stop it first and ensure /tmp/hadoop-saumya-namenode.pid file is empty before retry.
Starting datanodes
localhost: datanode is running as process 874. Stop it first and ensure /tmp/hadoop-saumya-datanode.pid file is empty before retry.
Starting secondary namenodes [LAPTOP-B593LG0J]
LAPTOP-B593LG0J: secondarynamenode is running as process 1100. Stop it first and ensure /tmp/hadoop-saumya-secondarynamenode.pid file is empty before retry
.
Starting resourcemanager
resourcemanager is running as process 1384. Stop it first and ensure /tmp/hadoop-saumya-resourcemanager.pid file is empty before retry.
Starting nodemanagers
localhost: nodemanager is running as process 1501. Stop it first and ensure /tmp/hadoop-saumya-nodemanager.pid file is empty before retry.
saumya@LAPTOP-B593LG0J:~$ jps
2550 Jps
1384 ResourceManager
874 DataNode
1100 SecondaryNameNode
764 NameNode
1501 NodeManager
saumya@LAPTOP-B593LG0J:~$ pwd
/home/saumya
saumya@LAPTOP-B593LG0J:~$ ls
hadoop-3.3.6  hadoop-3.3.6.tar.gz  hadoopdata
saumya@LAPTOP-B593LG0J:~$ pwd
/home/saumya
saumya@LAPTOP-B593LG0J:~$ java -version
java version "11.0.31" 2026-04-21
saumya@LAPTOP-B593LG0J:~$ hadoop version
OpenJDK Runtime Environment (build 11.0.31+11-post-lubuntu1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 11.0.31+11-post-lubuntu1-26.04.2-Ubuntu, mixed mode, sharing)
Hadoop 3.3.6
Source code repository https://github.com/apache/hadoop.git -r 1be78238728da9266a4f88195058f88fd012bf9c
Compiled by ubuntu on 2023-06-18T08:22Z
Compiled on platform linux-x86_64
```

```
saumya@LAPTOP-B593LG0J: ~$ hdfs dfsadmin -report
764 NameNode
1501 NodeManager
saumya@LAPTOP-B593LG0J:~$ hdfs dfsadmin -report
Configured Capacity: 1081101176832 (1006.85 GB)
Present Capacity: 1021958672384 (951.77 GB)
DFS Remaining: 1021958639616 (951.77 GB)
DFS Used: 32768 (32 KB)
DFS Used%: 0.00%
Replicated Blocks:
  Under replicated blocks: 0
  Blocks with corrupt replicas: 0
  Missing blocks: 0
  Missing blocks (with replication factor 1): 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0
Erasure Coded Block Groups:
  Low redundancy block groups: 0
  Block groups with corrupt internal blocks: 0
  Missing block groups: 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0
-----
Live datanodes (1):
Name: 127.0.0.1:9866 (localhost)
Hostname: LAPTOP-B593LG0J.localdomain
Decommission Status : Normal
Configured Capacity: 1081101176832 (1006.85 GB)
DFS Used: 32768 (32 KB)
Non DFS Used: 4158149120 (3.87 GB)
DFS Remaining: 1021958639616 (951.77 GB)
DFS Used%: 0.00%
DFS Remaining%: 94.53%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Thu Jun 11 02:53:54 UTC 2026
```

5. Conclusion

Hadoop was successfully installed and configured in a single-node environment. HDFS and YARN services were started and verified using JPS and web interfaces.

Experiment 2: Hadoop Basic Commands

Title: Hadoop Basic Commands

1. Objective

To understand and execute basic Hadoop HDFS commands for file and directory management.

2. Procedure

- Create a directory in HDFS.
- Upload files to HDFS.
- List files and directories.
- View file contents.
- Download files from HDFS.
- Remove files and directories.

3. Source Code

➤ Create directory

```
hdfs dfs -mkdir /demo
```

➤ Upload file

```
hdfs dfs -put sample.txt /demo
```

➤ List files

```
hdfs dfs -ls /demo
```

➤ View file content

```
hdfs dfs -cat /demo/sample.txt
```

➤ Download file

```
hdfs dfs -get /demo/sample.txt
```

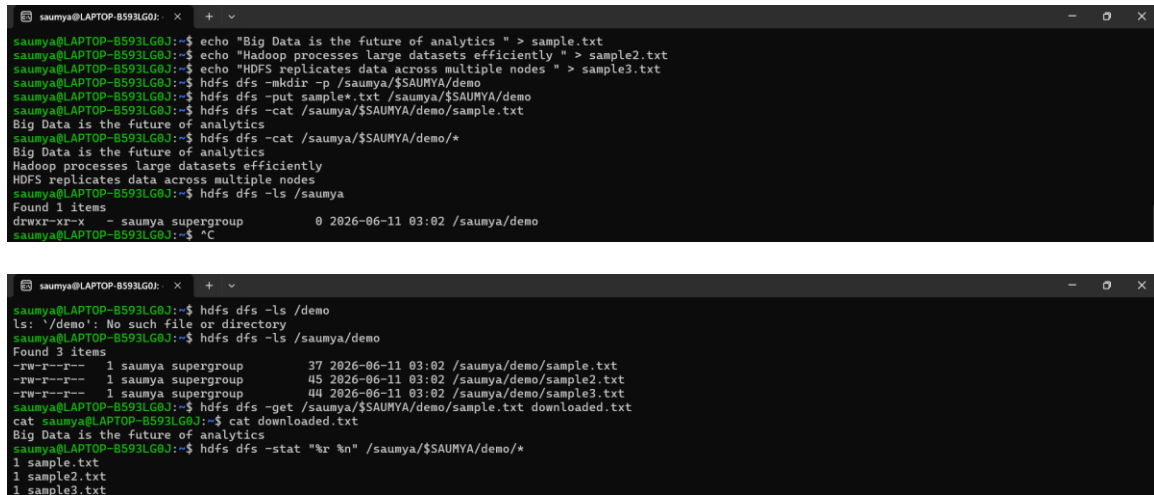
➤ Delete file

```
hdfs dfs -rm /demo/sample.txt
```

➤ Delete directory

```
hdfs dfs -rmdir /demo
```

4. Screen Shot



```
saumya@LAPTOP-B593LG0J:~$ echo "Big Data is the future of analytics " > sample.txt
saumya@LAPTOP-B593LG0J:~$ echo "Hadoop processes large datasets efficiently " > sample2.txt
saumya@LAPTOP-B593LG0J:~$ echo "HDFS replicates data across multiple nodes " > sample3.txt
saumya@LAPTOP-B593LG0J:~$ hdfs dfs -mkdir -p /saumya/$SAUMYA/demo
saumya@LAPTOP-B593LG0J:~$ hdfs dfs -put sample*.txt /saumya/$SAUMYA/demo
saumya@LAPTOP-B593LG0J:~$ hdfs dfs -cat /saumya/$SAUMYA/demo/sample.txt
Big Data is the future of analytics
saumya@LAPTOP-B593LG0J:~$ hdfs dfs -cat /saumya/$SAUMYA/demo/*
Big Data is the future of analytics
Hadoop processes large datasets efficiently
HDFS replicates data across multiple nodes
saumya@LAPTOP-B593LG0J:~$ hdfs dfs -ls /saumya
Found 1 items
drwxr-xr-x  - saumya supergroup          0 2026-06-11 03:02 /saumya/demo
saumya@LAPTOP-B593LG0J:~$ ^C

saumya@LAPTOP-B593LG0J:~$ hdfs dfs -ls /demo
ls: '/demo': No such file or directory
saumya@LAPTOP-B593LG0J:~$ hdfs dfs -ls /saumya/demo
Found 3 items
-rw-r--r--  1 saumya supergroup          37 2026-06-11 03:02 /saumya/demo/sample.txt
-rw-r--r--  1 saumya supergroup          45 2026-06-11 03:02 /saumya/demo/sample2.txt
-rw-r--r--  1 saumya supergroup          44 2026-06-11 03:02 /saumya/demo/sample3.txt
saumya@LAPTOP-B593LG0J:~$ hdfs dfs -get /saumya/$SAUMYA/demo/sample.txt downloaded.txt
cat saumya@LAPTOP-B593LG0J:~$ cat downloaded.txt
Big Data is the future of analytics
saumya@LAPTOP-B593LG0J:~$ hdfs dfs -stat "%r %n" /saumya/$SAUMYA/demo/*
1 sample.txt
1 sample2.txt
1 sample3.txt
```

5. Conclusion

Basic HDFS commands were executed successfully. Files and directories were created, uploaded, viewed, and deleted using Hadoop commands.

Experiment 3: Simple Word Count Program in Hadoop

Title: Simple Word Count Program in Hadoop

1. Objective

To perform word frequency analysis on a text file using the Hadoop MapReduce WordCount program.

2. Procedure

- Create an input text file.
- Upload the file to HDFS.
- Execute the Hadoop WordCount MapReduce job.
- View the generated output.
- Analyze the word frequency results.

3. Source Code

➤ Create input file

```
echo "Big Data is the future of analytics" > sample.txt
```

➤ Create HDFS input directory

```
hdfs dfs -mkdir /input
```

➤ Upload file

```
hdfs dfs -put sample.txt /input
```

➤ Run WordCount

```
hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar wordcount /input /output
```

➤ View results

```
hdfs dfs -cat /output/part-r-00000
```

4. Screen Shot

```
saumya@LAPTOP-B593LG0J: ~$ cat > shakespeare.txt
EOF
saumya@LAPTOP-B593LG0J:~$ cat shakespeare.txt
cat: shakespeare.txt: No such file or directory
saumya@LAPTOP-B593LG0J:~$ cat > shakespeare.txt << 'EOF'
To be, or not to be, that is the question.
All the world's a stage, and all the men and women merely players.
Some are born great, some achieve greatness.
EOF
saumya@LAPTOP-B593LG0J:~$ cat shakespeare.txt
To be, or not to be, that is the question.
All the world's a stage, and all the men and women merely players.
Some are born great, some achieve greatness.
saumya@LAPTOP-B593LG0J:~$ hdfs dfs -p /saumya/$SAUMYA/wordcount/input
^[[A*[[A-p: Unknown command
Usage: hadoop fs [generic options]
    [-appendToFile [-n] <localsrc> ... <dst>]
    [-cat [-ignoreCrc] <src> ...]
    [-checksum [-v] <src> ...]
    [-chgrp [-R] GROUP PATH...]
    [-chmod [-R] <MODE[,MODE]... | OCTALMODE> PATH...]
    [-chown [-R] [OWNER][:[GROUP]] PATH...]
    [-concat <target path> <src path> <src path> ...]
    [-copyFromLocal [-f] [-p] [-l] [-d] [-t <thread count>] [-q <thread pool queue size>] <localsrc> ... <dst>]
    [-copyToLocal [-f] [-p] [-crc] [-ignoreCrc] [-t <thread count>] [-q <thread pool queue size>] <src> ... <localdst>]
    [-count [-q] [-h] [-v] [-t <storage type>]] [-u] [-x] [-e] [-s] <path> ...]
    [-cp [-f] [-p] [-p[topax]] [-d] [-t <thread count>] [-q <thread pool queue size>] <src> ... <dst>]
    [-createSnapshot <snapshotDir> [<snapshotName>]]
    [-deleteSnapshot <snapshotDir> [<snapshotName>]]
    [-df [-h] [<path> ...]]
    [-du [-s] [-h] [-v] [-t] <path> ...]
    [-expunge [-immediate] [-fs <path>]]
    [-find <path> ... <expression> ...]
    [-get [-f] [-p] [-crc] [-ignoreCrc] [-t <thread count>] [-q <thread pool queue size>] <src> ... <localdst>]
    [-getfacl [-R] <path>]
    [-getfattr [-R] [-n name | -d] [-e en] <path>]
    [-getmerge [-n] [-skip-empty-file] <src> <localdst>]
    [-head <file>]
    [-help [cmd ...]]
    [-ls [-C] [-d] [-h] [-q] [-R] [-t] [-S] [-r] [-u] [-e] [<path> ...]]
```

```
saumya@LAPTOP-B593LG0J:~$ hdfs dfs -mkdir -p /saumya/$SAUMYA/wordcount/input
saumya@LAPTOP-B593LG0J:~$ hdfs dfs -put shakespeare.txt /saumya/$SAUMYA/wordcount/input/
saumya@LAPTOP-B593LG0J:~$ hadoop jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-*.jar \
> wordcount /saumya/$SAUMYA/wordcount/input /saumya/$SAUMYA/wordcount/output
2026-06-11 04:07:38,574 INFO client.DefaultHARFollowerProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-06-11 04:07:39,564 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/saumya/.staging/job_1781145962703_00
01
2026-06-11 04:07:40,225 INFO input.FileInputFormat: Total input files to process : 1
2026-06-11 04:07:40,367 INFO mapreduce.JobSubmitter: number of splits:1
2026-06-11 04:07:40,794 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1781145962703_0001
2026-06-11 04:07:40,794 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-06-11 04:07:41,189 INFO conf.Configuration: resource-types.xml not found
2026-06-11 04:07:41,110 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-06-11 04:07:42,082 INFO impl.YarnClientImpl: Submitted application application_1781145962703_0001
2026-06-11 04:07:42,175 INFO mapreduce.Job: The url to track the job: http://LAPTOP-B593LG0J.localdomain:8088/proxy/application_1781145962703_0001/
2026-06-11 04:07:42,177 INFO mapreduce.Job: Running job: job_1781145962703_0001
2026-06-11 04:07:56,494 INFO mapreduce.Job: Job job_1781145962703_0001 running in uber mode : false
2026-06-11 04:07:56,495 INFO mapreduce.Job: map 0% reduce 0%
2026-06-11 04:08:03,600 INFO mapreduce.Job: map 100% reduce 0%
2026-06-11 04:08:11,726 INFO mapreduce.Job: map 100% reduce 100%
2026-06-11 04:08:11,745 INFO mapreduce.Job: Job job_1781145962703_0001 completed successfully
2026-06-11 04:08:11,880 INFO mapreduce.Job: Counters: 54
File System Counters
  FILE: Number of bytes read=301
  FILE: Number of bytes written=552691
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=280
  HDFS: Number of bytes written=191
  HDFS: Number of read operations=8
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Launched map tasks=1
  Launched reduce tasks=1
  Data-local map tasks=1
  Total time spent by all maps in occupied slots (ms)=4754
  Total time spent by all reduces in occupied slots (ms)=4917
```

```
saumya@LAPTOP-B593LG0J: x + v
WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=155
File Output Format Counters
  Bytes Written=191
saumya@LAPTOP-B593LG0J:~$ hdfs dfs -cat /saumya/$SAUMYA/wordcount/output/part-r-00000 | head -20
All 1
Some 1
To 1
a 1
achieve 1
all 1
and 2
are 1
be, 2
born 1
great, 1
greatness. 1
is 1
men 1
merely 1
not 1
or 1
players. 1
question. 1
some 1
saumya@LAPTOP-B593LG0J:~$ hdfs dfs -ls /saumya/$SAUMYA/wordcount/output/
Found 2 items
-rw-r--r-- 1 saumya supergroup 0 2026-06-11 04:08 /saumya/wordcount/output/_SUCCESS
-rw-r--r-- 1 saumya supergroup 191 2026-06-11 04:08 /saumya/wordcount/output/part-r-00000
saumya@LAPTOP-B593LG0J:~$
```

5. Conclusion

The Hadoop MapReduce WordCount program was executed successfully. The output displayed the frequency of each word in the input file, demonstrating the distributed processing capability of Hadoop.